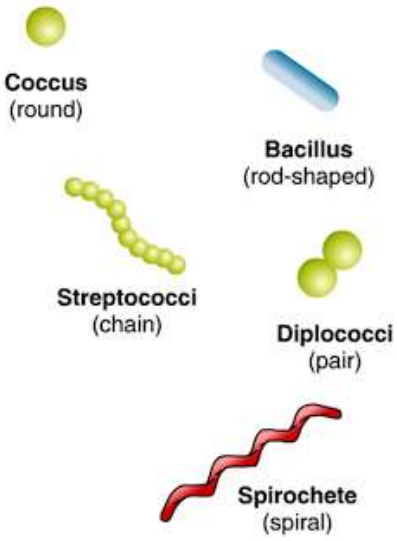
CLASSIFICATION OF MICROORGANISMS

CLASSIFICATION OF MICROORGANISMS

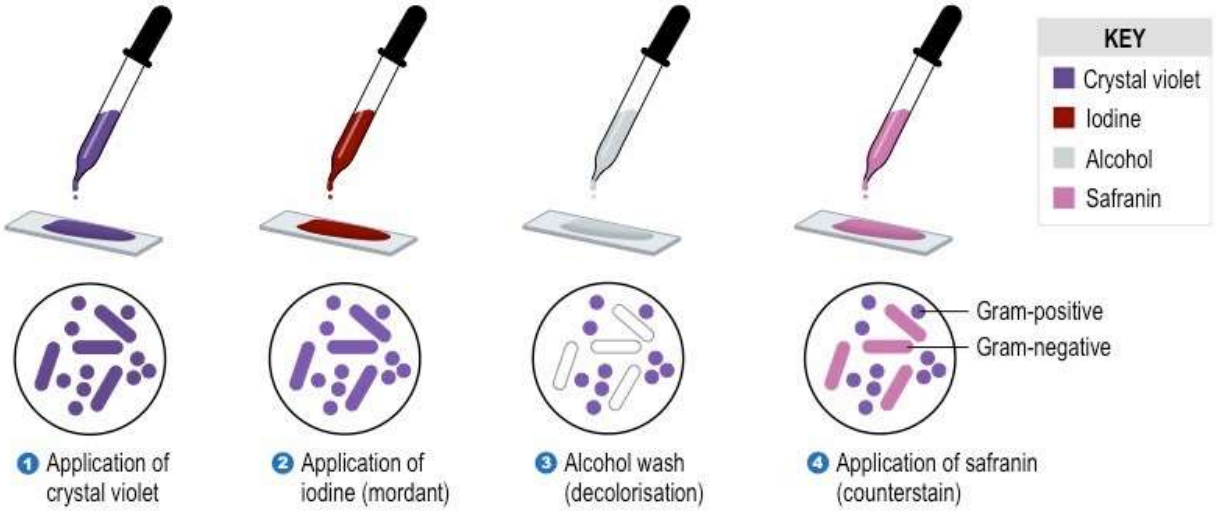
- All the organisms on the earth are divided into 5 kingdoms: Monera, Protista, Fungi, Animalia, and Plantae
- Microorganisms are included in the first 3 kingdoms
- All the prokaryotic cells i.e. bacteria are included in Monera
- Protozoans, algae, slime molds, and water molds are included in the kingdom Protista
- Fungi are included in the kingdom Fungi

CLASSIFICATION OF BACTERIA

- All the bacteria or prokaryotic organisms are included in the kingdom Monera and they can be further **classified on the basis of many different characteristics**
- **Bacteria** are small unicellular organisms with no defined nucleus, membrane-bound cell organelles that divide by binary fission
- According to the American Society for Microbiology, bacteria are one of the most diverse life forms on earth and may consist of more than one million species
- The simplest of classifications is the one based on the shape of the bacteria
- **Based on the shape**, the bacteria are classified into
    - Cocci – spherical shaped
    - Diplococci – pairs of cocci
    - Streptococci – chains of cocci
    - Staphylococci – bunches of cocci
    - Bacillus – rod shaped
    - Coccobacilli – with a shape that is intermediate between cocci and bacilli i.e. very short rods or ovals
    - Streptobacilli – chain of bacilli
    - Comma shaped – best example is *Vibrio*
    - Spirillum – spiral shaped



- They can also be classified based on the phenotypic characteristics for instance based on the structure of the cell wall which can be easily determined and forms the basis for Gram staining
- So, they can be **classified on the basis of the Gram Staining**
    - **Gram-positive**: all the bacteria that appear purple/blue under microscope after Gram staining (steps involved in Gram staining can be seen the image below)
    - **Gram-negative**: all the bacteria that appear pink/red under microscope after Gram staining

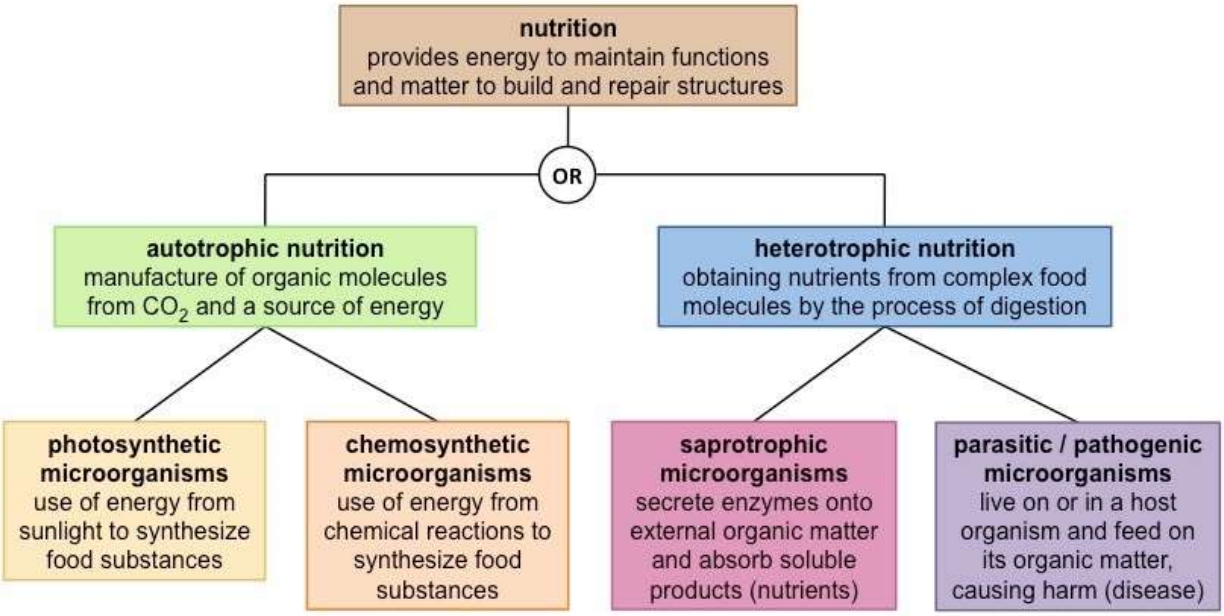


- Another classification is **based on the growth requirements or mode of nutrition**
    - Broadly divided into two groups, Autotrophs and Heterotrophs
    - **Autotrophs** – bacteria that use carbon dioxide as the sole source of carbon and generate organic molecules necessary for their growth. Autotrophs need energy in

order to generate these organic molecules from Carbon dioxide. Depending on the source of energy used, they are further divided into Phototrophs and Chemotrophs.

- Phototrophs – they gain energy by using light. These are further divided into 2 types depending on the source of their electrons; Photolithotrophs use inorganic compounds like hydrogen sulfide as electron source while photoorganotrophs use organic compounds like succinate as the electron source
- Chemoautotrophs – these bacteria obtain their energy using chemical compounds. They are further divided into 2 groups based on the electron source; chemolithotrophs which gain energy from inorganic compounds like ammonia as electron source while chemoorganotrophs use organic compounds like glucose or amino acids as electron source

- **Heterotrophs** – bacteria that use reduced or pre-formed organic molecules from other organisms as their carbon source for growth. They can be further divided into parasites and saprophytes. While parasites feed on live organisms, saprophytes feed on dead and decaying matter



- They can also **be classified based on their oxygen requirements** for growth
  - **Obligate aerobes** – oxygen is must for growth
  - **Facultative anaerobes** – can grow with or without the presence of oxygen
  - **Obligate anaerobes** – cannot survive in the presence of oxygen
  - **Microaerophilic** – need low concentrations of oxygen for growth

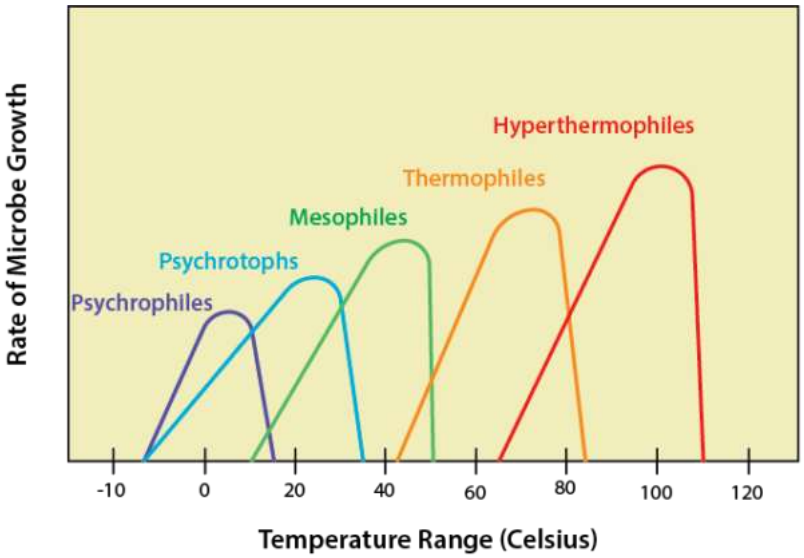
Gram Positive Bacteria

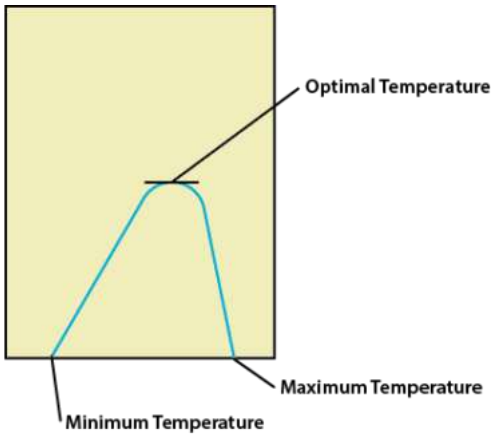
Name	Morphology	O_2 Requirements	Commensal
Staphylococci	Cocci in grape-like clusters	facultative anaerobe	Yes
Streptococci	Cocci in pairs, chains	facultative anaerobe	Some species
Pneumococci	Diplococci, lancet shaped	facultative anaerobe	±
Enterococci	Cocci in pairs, chains	facultative anaerobe	Yes
Bacilli	Rods, spore-forming	aerobic	±
Clostridia	Rods, spore formers	anaerobic	Some species
Corynebacterium	Rods, nonspore forming	facultative anaerobe	Some species
Listeria	Rods, nonspore formers	facultative anaerobe	No
Actinomyces	Irregular, filamentous, form sulfur granules	anaerobic	Yes

Gram Negative Bacteria

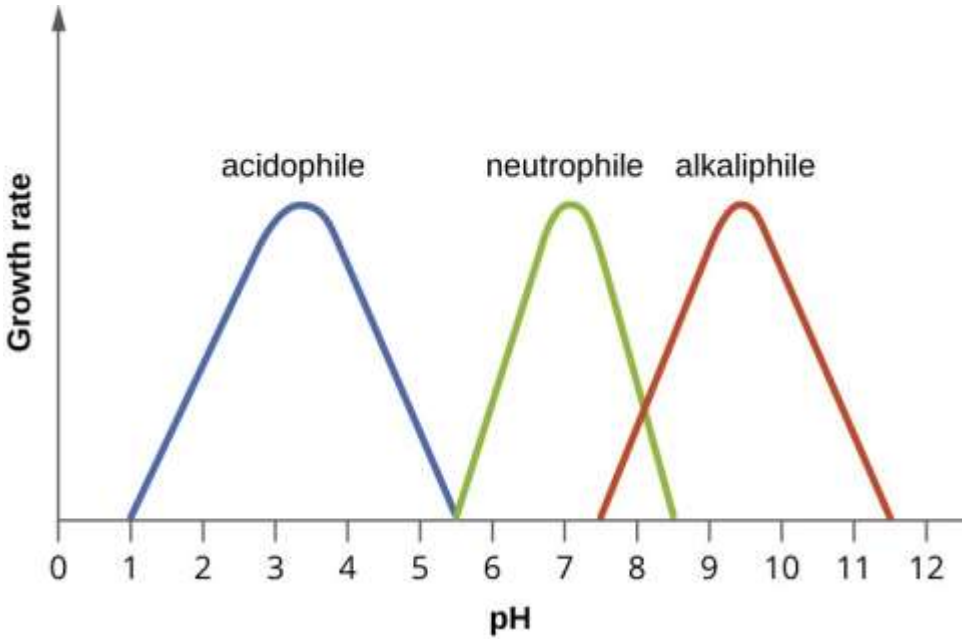
Name	Morphology	O_2 Requirements
Enterobacteriaceae (*E. coli*, klebsiella, salmonella, shigella)	Rods	facultative anaerobe
Bacteroides	Rods	anaerobic
Pseudomonas	Rods	aerobic
Vibrio (cholera)	Rods, curved shape	microaerophilic
Campylobacter	Rods, curved shape	microaerophilic
Legionella	Rods, poorly stained	microaerophilic
Neisseria	Cocci, kidney-bean shaped	Microaerophilic
Hemophilus	Coccobacillary - pleomorphic	facultative anaerobe
Bartonella	Small, pleomorphic rods	aerobic / microaerophilic

- Can also be **classified on the basis of optimum temperature required for the growth**, the bacteria are classified into
    - **Psychrophiles** – bacteria which grow at temperatures of 15 °C or lower
    - **Psychrotrophs** - cold-tolerant bacteria or Archaea that can grow at low temperatures, but have optimal and maximal growth temperatures above 15 and 20°C.
    - **Mesophiles** – bacteria with optimum growth at temperatures between 25-40 °C
    - **Thermophiles** – bacteria that show optimum growth above 45 °C
    - **Hyperthermophiles** – bacteria that grow at extreme temperatures i.e. above 80 °C

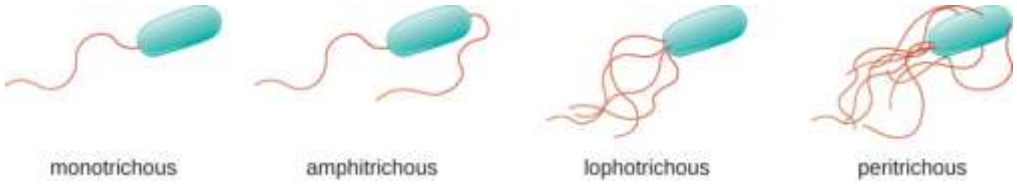




- Bacteria can also **be classified on the basis of pH required for their growth**
    - **Acidophiles** – grow best at acidic pH
    - **Alkaliphiles** – grow best at basic pH
    - **Neutrophiles** – grow best at neutral pH (6.5 to 7.5). Most of the pathogenic bacteria of humans are neutrophiles



- Bacteria can also **be classified on the basis of the flagella arrangement**
    - **Monotrichous** - a single flagellum at one pole (also called polar flagellum) example- *Vibrio cholerae*
    - **Amphitrichous** - single flagellum at both poles. Example- *Spirilla*
    - **Lophotrichous** - two or more flagella at one or both poles of the cell. Example- *Spirillum undula*
    - **Peritrichous** - completely surrounded by flagella. Example- *E. coli.*



- Another important classification that is extensively used is the **Bergey's Classification**.
    - David Bergey a Professor of Bacteriology published manuals for the classification of bacteria
    - In his manual of systemic bacteriology (**Bergey's Manual of Systemic Bacteriology**), the basis of classification is the evolutionary or genetic relationships between the bacteria
    - In **Bergey's Manual of Determinative Bacteriology**, the classification is based on several different criteria like the cell wall composition, morphology, biochemical tests, differential staining etc.

Important notes: A lot of questions have been asked based on the various types of classification of bacteria. It is important to not only to understand and remember the various terms but also important to memorize the various examples for each type of classification

CLASSIFICATION OF ARCHAEA

- Archaea differ from bacteria in many ways
    - They have distinctive rRNA sequences
    - They lack peptidoglycan in their cell walls and have unique membrane lipids
    - They are found in extreme environment conditions like high temperatures, salt concentrations etc.
- Archae bacteria are classified into 3 main types
    - Methanogens – occur in marshy areas and have the ability to convert formic acid and carbon dioxide to methane using hydrogen
    - Halophiles – occur in salt rich environments like sea beds, salt beds, salt marshes etc.
    - Thermoacidophiles – include bacteria that can withstand high temperatures and high acidity. They can often be seen in hot sulphur springs

CLASSIFICATION OF EUKARYOTIC MICROORGANISMS

- Eukaryotic microorganisms are broadly divided into 2 categories – protists and fungi
- Protists are unicellular like the bacteria but are larger when compared to bacteria
- Protists include protozoans, algae, slime molds, and water molds
    - Algae – photosynthetic; prepare all their food using sunlight
    - Protozoa – unicellular, usually motile, can be free living or pathogenic. Some of the protists like *Entamoeba, Plasmodium* are pathogenic to human beings
    - Slime molds – behave like protozoans in one stage of their life cycle and like fungi in the other
    - Water molds – grow on surface of fresh water and moist soil
- Unlike protists, fungi are a very diverse group of organisms with both unicellular structure (ex: yeast) or multicellular structure (ex: mushrooms)
- Fungi are heterotrophs some of which have the ability to grow on dead and decomposing organisms
- Some fungi associate with the roots of plants to form mycorrhizae
- There are many fungi that are pathogenic to humans and many different plants

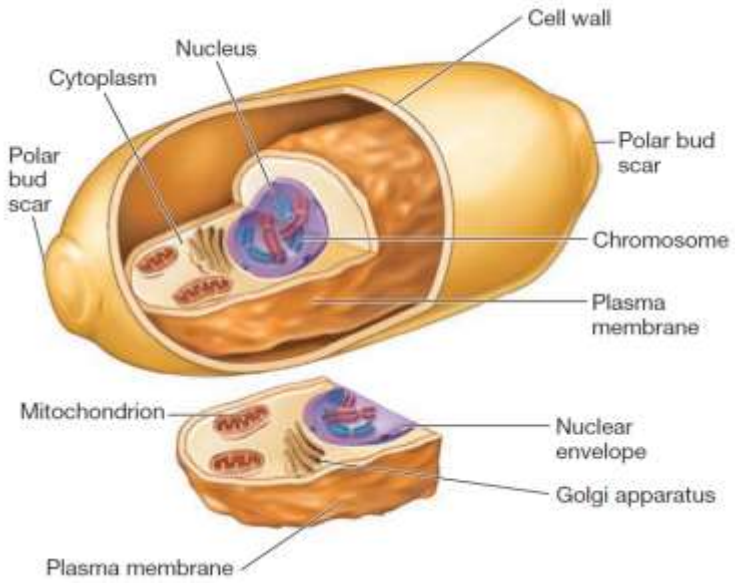
Yeast and Molds (Fungi)

- Microbiologists use the term fungus [pl., fungi] to describe eukaryotic organisms that are spore-bearing, have absorptive nutrition, lack chlorophyll, and reproduce sexually and asexually.
- Scientists who study fungi are **mycologists** and the scientific discipline devoted to fungi is called **mycology**.
- The study of fungal toxins and their effects is called **mycotoxicology**, and the diseases caused by fungi in animals are known as **mycoses** (s., mycosis).
- Fungi are primarily **terrestrial** organisms, although a few are freshwater or marine.
- They have a global distribution from polar to tropical regions.
- Many are pathogenic and infect plants and animals.
- Fungi also form beneficial relationships with other organisms.
- Fungi also are found in the upper portions of many plants, called **endophytic fungi**.
- These endophytic fungi affect plant reproduction and palatability to herbivores.
- **Lichens** are associations of fungi and photosynthetic protists or cyanobacteria.

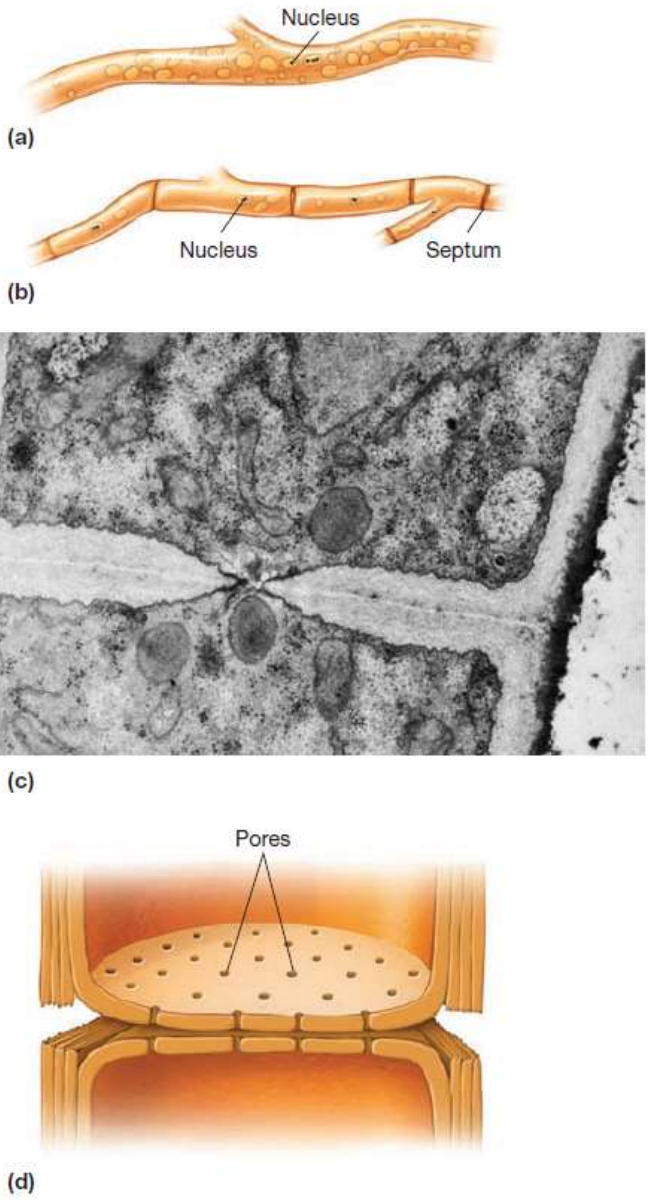
GENERAL FEATURES

- The body or vegetative structure of a fungus is called a **thallus** [pl., thalli].
- It varies in complexity and size, ranging from the single-cell microscopic yeasts to multicellular molds, macroscopic puffballs, and mushrooms.
- The fungal cell usually is encased in a cell wall of **chitin**.
- Chitin is a strong but flexible nitrogen containing polysaccharide consisting of **N-acetylglucosamine** residues.
- A yeast is a unicellular fungus that has a single nucleus and reproduces either asexually by budding or sexually through spore formation.
- Each bud that separates can grow into a new yeast, and some group together to form colonies.

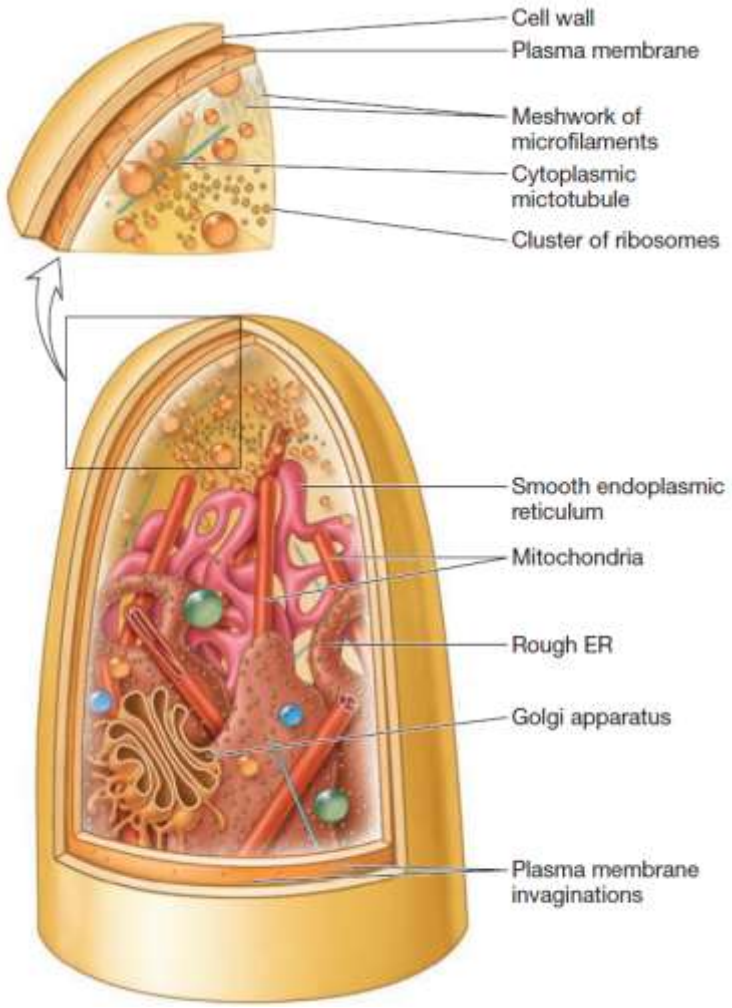
- Generally yeast cells are larger than bacteria, vary considerably in size, and are commonly spherical to egg shaped.
- They lack flagella but possess most of the other eukaryotic organelles
- The thallus of a mold consists of long, branched, threadlike filaments of cells called **hyphae** [s., hypha]
- The aggregation of hyphae results in the formation of **mycelium** (pl., mycelia), a tangled mass or tissue.
- In some fungi, protoplasm streams through hyphae and such hyphae are called **coenocytic or aseptate**.
- The hyphae of other fungi have interruptions called **septa** (s., septum) with either a single pore or multiple pores that enable cytoplasmic streaming. These hyphae are termed **septate**.
- Hyphae are composed of an outer cell wall and an inner **lumen**, which contains the cytosol and organelles
- A plasma membrane surrounds the cytoplasm and lies adjacent to the cell wall.
- The filamentous nature of hyphae results in a large surface area relative to the volume of cytoplasm, to make adequate nutrient absorption possible.
- Many fungi, especially those that cause diseases in humans and animals, are **dimorphic,** that is, they have two forms.
- Dimorphic fungi can change from the yeast (Y) form to the mold or mycelial form (M) in response to changes in various environmental factors This shift is called the **YM shift**.



Diagrammatic drawing of a yeast cell showing typical morphology. For clarity, the plasma membrane has been drawn separated from the cell wall. In a living cell the plasma membrane adheres tightly to the cell wall.



Drawings of (a) coenocytic hyphae (aseptate) and (b) hyphae divided into cells by septa. (c) Electron micrograph of a section of *Drechslera sorokiniana* showing wall differentiation and a single pore. (d) Drawing of a multiperforated septal structure.



Hyphal Morphology. Diagrammatic representation of a hyphal tip showing typical organelles and other structures.

Important notes: 1. Fungi includes Unicellular yeasts as well as multicellular hyphae containing molds. 2. Their cell wall contains Chitin 3. They are Heterotrophs

PROTISTS (PROTOZOANS)

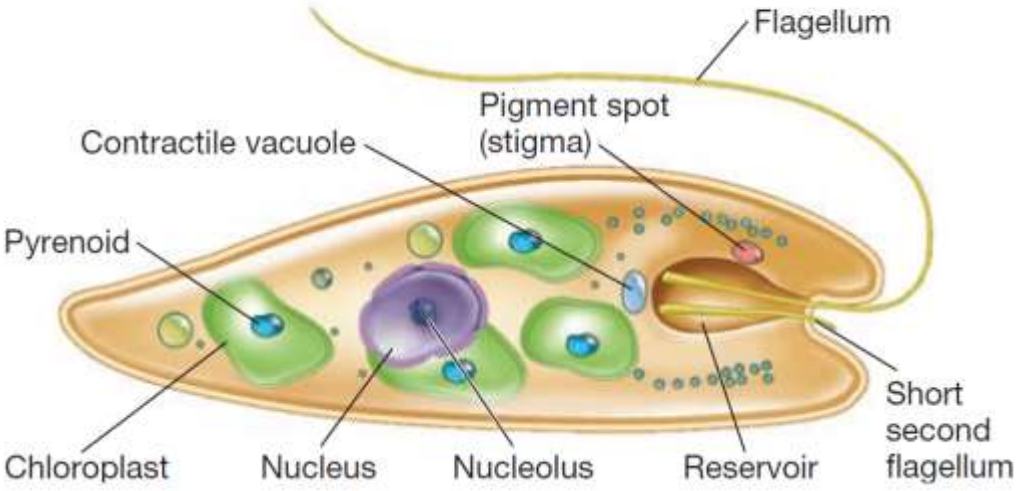
- The kingdom **Protista**, as defined by Whittaker's five-kingdom scheme, is an artificial grouping of over 64,000 different single-celled life forms that lack common evolutionary origin, are termed **polyphyletic**.

- In fact, the protists lack the level of tissue organization found in fungi, plants, and animals.
- The term **protozoa** [s., protozoan] is referred to as chemoorganotrophic protists, and **protozoology** generally refers to the study of protozoa.
- The term **algae** describe photosynthetic protists and originally used to refer to all "simple aquatic plants," earlier.
- The study of photosynthetic protists (algae) is often referred to as **phycology** and is of interest to both **botanists** and **protistologists**.
- The study of all protists, regardless of their metabolic type, is called **protistology**.
- The protozoa are classified into four major groups based on their means of locomotion: **flagellates** (Mastigophora), **ciliates** (Infusoria or Ciliophora), **amoebae** (Sarcodina), and **stationary forms** (Sporozoa).

GENERAL FEATURES

- Protists grow in a wide variety of moist habitats.
- Moisture is absolutely necessary for their existence because they are susceptible to desiccation.
- Most protists are free living and inhabit freshwater or marine environments.
- Many terrestrial chemoorganotrophic forms can be found in decaying organic matter and in soil, where they recycle essential elements, nitrogen and phosphorus.
- Other forms are **planktonic**—floating free in lakes and oceans.
- Planktonic microbes are responsible for a majority of the nutrient cycling that occurs in these ecosystems.
- Every major group of protists live in association with other organisms.
- Because protists are eukaryotic cells, their morphology and physiology are the same as the cells of multicellular plants and animals.
- Protists are small in size, that results in high ratio of cell surface to intracellular volume, to facilitate the exchange of nutrients and metabolites

- The largest algae (the **seaweeds**) are long, thin, and often flattened, in order to maximize the surface-to-volume ratio.
- The protist cell membrane is called **plasmalemma**, identical to multicellular organisms.
- The cytoplasm is present under the plasmalemma and is divided into an outer gelatinous region called the **ectoplasm**, and an inner fluid region, the **endoplasm**.
- The ectoplasm imparts rigidity to the cell body.
- The plasmalemma and structures present beneath it are called the **pellicle**.
- One or more vacuoles are usually present in the cytoplasm of protozoa.
- **Contractile vacuoles** function as osmoregulatory organelles in protists, living in hypotonic environments, such as freshwater lakes.
- The majority of **anaerobic protists** (such as Trichonympha, which lives in the gut of termites) lack mitochondria and cytochromes, and have an incomplete tricarboxylic acid cycle.
- Some protists have small, membrane-bound organelles termed **hydrogenosomes**.
- **Photosynthetic protists** have chloroplasts featuring **thylakoid membranes**.
- A dense proteinaceous area, called **pyrenoid**, is associated with the synthesis and storage of starch.
- Many protists contain cilia or flagella at some point in their life cycle.
- Their formation is associated with a basal body-like organelle called the **kinetosome**
- These organelles provide motility and are used to generate water currents for feeding and respiration.



Euglena. A Diagram Illustrating the Principle Structures Found in this Euglenoid.

VIRUSES AND THEIR CLASSIFICATION

- Viruses are acellular infectious agents that can multiply only in a host cell
- They are usually composed of a nucleic acid, some proteins and some may even have a lipid containing membrane called the envelope
- Viruses are classified on the basis of many different characteristics. Some of the most important classifications include classification based on the type of nucleic acid present also called as Baltimore classification
    - Group I – contain double stranded DNA
    - Group II – contain single stranded DNA
    - Group III – contain double stranded RNA
    - Group IV – contain single stranded RNA with positive strands (positive polarity)
    - Group V – contain single stranded RNA with negative strands (negative polarity)
    - Group VI - contain single stranded RNA which is associated with enzyme reverse transcriptase
    - Group VII - contain double stranded DNA which is associated with enzyme reverse transcriptase
- They are also classified into enveloped and non-enveloped viruses based on the presence or absence of the envelope respectively
- On the basis of the type of host they invade, viruses are classified into
    - Animal viruses – ex: polio virus
    - Plant viruses – ex: TMV
    - Bacteriophages – ex: viruses that infect bacteria
- Another classification of viruses is based on the symmetry and the shape of the capsid
- The major classes based on capsid symmetry include – helical (rod-like), icosahedral (spear-like)

- Based on the shape, viruses are classified into
    - rabies virus – bullet shaped
    - pox virus – brick shaped
    - ebola virus – filamentous shaped
    - adenovirus – space vehicle shaped

Class	Method of replication	Example
I	double- stranded DNA viruses	*Adenoviruses, Herpesviruses, Poxviruses*
II	single-stranded DNA viruses	*Parvoviruses*
III	double- stranded RNA viruses	*Reoviruses*
IV	single-stranded positive sense RNA viruses	*Picornaviruses, Togaviruses*
V	single-stranded negative sense RNA viruses	*Orthomyxoviruses, Rhabdoviruses*
VI	positive-sense single-stranded RNA viruses that replicate through a DNA intermediate	*Retroviruses*
VII	double-stranded DNA viruses that replicate through a single stranded RNA intermediate	*Hepadnaviruses*

Important notes: The most important thing in the viruses' part is the genome that they contain. The kind of questions that are generally asked includes, e.g. Adenoviruses have _____ as their genome

MOLECULAR APPROACHES TO MICROBIAL TAXONOMY AND PHYLOGENY

- By definition, Taxonomy is the science of naming, describing and classifying organisms and includes all plants, animals and microorganisms of the world
- Phylogeny is the study of relationships among different groups of organisms and their evolutionary development.
- Both Taxonomy and Phylogeny initially started by sorting out the organisms based on phenotypic characteristics
- With the development of latest molecular techniques that allows sequencing the genome of the organisms led to a completely new approach of taxonomy and phylogeny
- With the advent of modern methods of sequencing, scientists started collecting genome information that allows them to make evolutionary connections between organisms and then do taxonomy and phylogeny based on the genome data rather than phenotypes
- The sequences that are chosen for the molecular phylogeny are the 16S rRNA gene sequences that are highly conserved between different species of bacteria and archaea
- The 16sRNA sequence data from many bacteria are collected and using appropriate bioinformatics tools phylogenic trees are generated which will provide a clear picture about the relationship between those organisms in terms of evolution